

Utilizing VLMs for Identifying Child Safety Hazards

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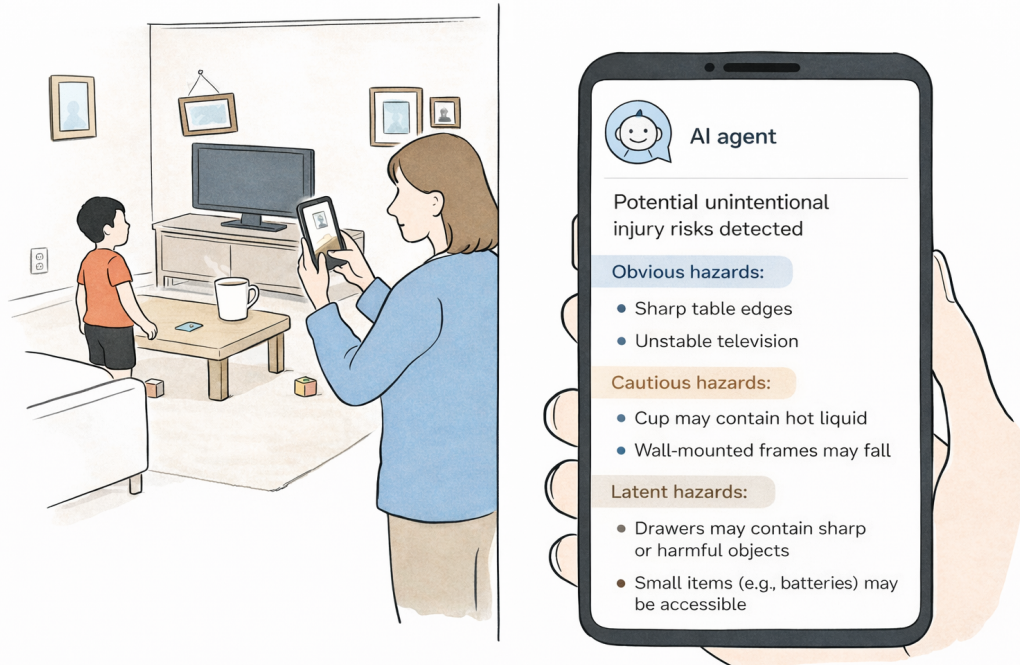


Figure 1: Conceptual illustration of an AI-assisted child safety scenario. A caregiver captures an image of a home environment, and an AI agent identifies potential unintentional injury risks.

Abstract

Unintentional injuries caused by everyday household hazards remain a leading threat to child safety, yet anticipating risk in dynamic home environments is cognitively demanding for caregivers. Recent vision-language models (VLMs) offer new opportunities for AI-assisted hazard identification by combining visual perception with contextual reasoning. In this work, we examine how a state-of-the-art VLM identifies child safety hazards in domestic environments and how its interpretations align with human expert judgments. We compared model- and human-identified hazards and found substantial overall agreement, particularly for visually salient risks. We also observed disagreements, with both humans and the VLM missing hazards that are low-salience, context-dependent, or require indirect inference, revealing complementary blind spots.

Our findings highlight the promise and limitations of VLMs for child safety and suggest that AI systems may best serve as complementary tools that surface overlooked risks rather than replacements for human judgment.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**; • **Computing methodologies** → **Computer vision**.

Keywords

Child Safety; Vision-Language Models; Hazard Detection; Human-AI Collaboration



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1 Introduction

Unintentional child injuries resulting from exposure to safety hazards such as falls, drowning, and suffocation, continue to be the leading cause of death among children aged between 1 to 4 years (CDC, 2015) [5], and the second leading cause of hospitalization (CDC, 2011) [5], in the United States (US). Among infants (<1 year), exposure to hazardous conditions in domestic environments represents a substantial risk, with unintentional injuries ranking as the third leading cause of death (CDC, 2021) [17]. During 2018-2019, an average of approximately 2,400 children aged 0–4 years died from unintentional injuries associated with preventable safety hazards, this represents ~7 child deaths per day that can be avoided. In addition to the emotional toll on the victims' families, these deaths are associated with substantial economic burden, exceeding \$14 billion annually in combined medical and work-loss costs (CDC, 2021) [17]. This problem is not just unique to the US as unintentional injuries rank among the top causes of deaths globally.

In response to this, contemporary approaches to child injury prevention largely emphasize parental supervision and in-home childproofing [17]. Within this framework, caregivers (particularly first-time parents) are expected not only to install prescribed safety devices, but also to anticipate potential hazards embedded in everyday domestic environments. While well-established guidelines exist for common childproofing measures [15], these recommendations implicitly assume relatively stable environments. In practice, however, homes are inherently dynamic, objects are moved, routines evolve, and new hazards emerge over time. Anticipating risk under these conditions is therefore not a one-time task, but an ongoing and cognitively demanding challenge for caregivers.

Recognizing these limitations, recent research has explored automated approaches for detecting and assessing safety hazards across a range of domains, including child and elder safety [1, 7], construction-site safety [8], and accessibility [14]. Most of these approaches leverage some form of supervised learning (often deep CNN or YOLO-based detectors) to recognize hazardous objects. The notion of hazard is usually defined by a fixed label set (e.g., knife), essentially framing risk as a predefined collection of objects or conditions. While this approach is effective for well-known dangers, it has important limitations. They tend to be narrow in scope, focusing on a limited taxonomy of hazards, and struggle with context-dependent or latent risks. For example, a standard object detector may successfully identify a knife placed on a countertop, yet fail to reason about the potential presence of a knife hidden inside a kitchen drawer.

Building on these advances, recent work has begun to explore vision-language models (VLMs) for safety-related reasoning. Unlike traditional convolutional neural network (CNN)-based pipelines, which focus on local visual features and offer limited support for relational reasoning, vision-language models leverage attention-based architectures [16] to enable more holistic interpretation of complex scenes. Prior studies show that these models can interpret images by considering how objects relate to one another and how context shapes potential risk, enabling applications in assistive robotics, accessibility, and safety analysis [2, 10]. Together, these capabilities suggest that VLMs may be particularly well suited for

identifying hazards that are not only immediately visible, but also latent in nature.

In this work, we investigate how VLMs identify hazards in child safety contexts, and how such model-identified hazards align with human interpretations of safety. To this end, we annotated images of everyday home environments. We then conducted a thematic analysis that identified three hazard categories: *obvious*, *cautious*, and *latent* (see subsection 3.2). Finally, we compared annotations generated by VLMs with those identified by human annotators to examine areas of alignment and divergence.

Our findings show that VLMs can identify many child safety hazards in domestic environments with substantial alignment to human expert judgment, particularly for hazards that are visually salient and widely recognized. At the same time, both humans and the model exhibit complementary blind spots, especially for hazards that are low-salience, context-dependent, or require indirect inference about child behavior. By characterizing hazards into different categories, our work provides a framework for understanding where current automated approaches succeed and where they fall short. Together, these insights motivate the use of VLMs not as replacements for caregiver judgment, but as complementary tools that surface overlooked risks and support reflective safety awareness in everyday home settings.

2 Related Work

2.1 Automated hazard detection

Prior work has explored automated approaches for detecting safety hazards in everyday environments using computer vision and machine learning. In domestic settings, researchers have trained image-based models to recognize common household hazards for children, such as sharp objects or accessible medications [1]. Related approaches have been developed for workplace safety, where object detection and pose estimation are used to identify personal protective equipment violations or unsafe postures on construction sites [7, 8]. Healthcare and assistive technologies similarly leverage vision-based activity recognition to detect falls and other safety-critical events among elderly users [3, 7].

While these approaches demonstrate the feasibility of automated hazard detection, they typically focus on visible, predefined hazards represented through fixed label sets. As a result, such systems often struggle to capture context-dependent or latent risks that emerge from object relationships or environmental conditions. Recent work has therefore begun to incorporate richer contextual reasoning into hazard assessment. For example, Rodríguez-Juan et al. combine object, action, and scene context models to assess multi-faceted risks in home videos [11]. Similarly, Ishii et al. introduce a graph-based representation that infers contextual hazards, such as when an otherwise benign object becomes dangerous due to its proximity to a child [6].

2.2 Vision-language models for holistic hazard understanding

Building on these developments, recent research has explored VLMs as a means of supporting hazard understanding in complex, dynamic environments. Unlike traditional single-task vision pipelines,

VLMs enable open-ended interpretation of scenes and the articulation of safety-relevant context [2, 9]. This capability allows such models to move beyond detecting potentially hazardous objects (e.g., a knife on a counter) toward reasoning about how risk emerges from context, access, and use.

Several initial studies have demonstrated the potential of VLMs for safety-related reasoning in domestic environments. For instance, VLM-powered robotic assistants have been shown to identify and describe household safety hazards, such as unattended stoves or unguarded pools, by reasoning over object relationships within a scene [10]. Related multimodal approaches have used generative models to produce diverse hypothetical hazard scenarios, improving a robot’s preparedness for rare or unexpected dangers [13]. Together, these findings suggest that integrating vision and language enables more robust hazard understanding, particularly in dynamic home environments where risks must be anticipated rather than simply detected.

3 Methods

3.1 Image Data Collection

We curated a small dataset of images depicting everyday home environments commonly encountered by children. The dataset consisted of 18 images representing bedrooms, bathrooms, living rooms, and kitchens, with at least four images per environment. All images were sourced from the Unsplash repository and selected to be freely usable without licensing restrictions.

3.2 Human Expert Annotation of Child Safety Hazards

Image annotation proceeded iteratively across multiple rounds until inter-annotator agreement reached approximately 80%. In each round, two human experts annotated child safety hazards independently to avoid influencing one another’s judgments. Disagreements (i.e., hazards identified by only one expert) were then identified. Experts subsequently discussed these cases to resolve ambiguous interpretations of what constitutes as a child safety hazard and to refine the shared annotation criteria. This process resulted in consistent hazard annotation guidelines that were applied across subsequent annotation sessions.

In the first round, five images were independently annotated, resulting in an inter-annotator agreement of 55%. Following discussion and refinement of the annotation criteria, a second round was conducted with five additional images. Agreement increased to 79% in this round, indicating improved alignment between annotators. A subsequent round with four additional images yielded the desired agreement of 80%, suggesting that the experts had developed a shared understanding of hazard identification. The remaining four images were evenly divided between the two experts and annotated independently using the finalized guidelines. After annotation was completed, experts consolidated the annotations by merging duplicate/repeated annotations within the same image.

3.3 Hazard Categorization

Following annotation, experts conducted an affinity diagramming process [4] to organize identified hazards and surface recurring

patterns in how risks were perceived and reasoned about across images. This analysis resulted in three high-level hazard categories: (1) *obvious*: hazards caused by objects visible in the image and widely recognized as dangerous (e.g., sharp table corners or exposed electrical outlets), *cautious*: hazards caused by objects visible in the image but whose risk depends on contextual factors (e.g., a cup that may contain hot liquid), and *latent*: hazards caused by objects not visible in the image but reasonably inferred from the environment (e.g., chemicals stored inside closed kitchen drawers, or outdoor access inferred from sunlight in the room). Each annotated hazard was subsequently assigned a category label based on this taxonomy.

3.4 Vision-language Model Prompting

Guided by the human expert annotation process, a structured prompt was developed to elicit child safety hazard interpretations from a VLM (GPT 5.2). The prompt was designed to encourage the model to reason beyond visible objects and consider contextual and inferred risks (i.e., latent hazards). Prompt wording was iteratively refined through expert discussion to align with the finalized annotation guidelines and the identified hazard categories. Each image was independently provided to the model using the same prompt, with no transfer of conversation history or contextual memory, to generate model-identified hazards. The prompt explicitly requested the model to output, for each identified hazard, a textual description, and an assigned hazard category label. These structured outputs enabled systematic comparison between VLM-generated annotations and human expert annotations.

4 Results

For each image, we examined both agreements and disagreements between human experts and the VLM manually, the results are summarized in Table 1. Agreement was defined semantically, with annotations considered a match if they referred to the same underlying hazard despite differences in wording between the humans and VLM. Across all images, a total of 352 hazard annotations were identified when combining human and VLM outputs, spanning the three hazard categories (*obvious*: 74, *cautious*: 188, *latent*: 90). Human experts identified a total of 201 hazards across all images (*obvious*: 30, *cautious*: 134, *latent*: 37). The VLM identified a total of 151 hazards (*obvious*: 44, *cautious*: 54, *latent*: 53).

4.1 Agreement Analysis

Across all annotations, 266 (75.6%) were in agreement between human experts and the VLM (*obvious*: 61, *cautious*: 142, *latent*: 63). Of these, 144 agreements corresponded to human-identified hazards (*obvious*: 24, *cautious*: 93, *latent*: 27), while 122 agreements corresponded to VLM-identified hazards (*obvious*: 37, *cautious*: 49, *latent*: 36). These counts are not in one-to-one correspondence, as multiple human annotations could semantically align with a single VLM-identified hazard, reflecting differences in annotation granularity. Agreement rates varied by hazard type, with the highest agreement observed for *obvious* hazards (82.4%; 61/74), followed by *cautious* hazards (75.5%; 142/188), and *latent* hazards (70.0%; 63/90).

Table 1: Summary of hazard agreement and disagreement between human experts and the VLM.

Metric	Obvious	Cautious	Latent	Total
All identified hazards	74	188	90	352
Agreed hazards (count (%))	61 (82.4%)	142 (75.5%)	63 (70.0%)	266 (75.6%)
Disagreed hazards (count (%))	13 (17.6%)	46 (24.5%)	27 (30.0%)	86 (24.4%)

4.2 Disagreement Analysis

Across all images, 86 of the 352 identified hazards (24.4%) reflected disagreements between human experts and the VLM (*obvious*: 13, *cautious*: 46, *latent*: 27). From the human perspective, experts identified 57 hazards that were missed by the VLM, the majority of which were *cautious* hazards (41/57), followed by *latent* (10/57) and *obvious* hazards (6/57). In contrast, the VLM identified 29 hazards that were not identified by human annotators, with these disagreements disproportionately concentrated in the *latent* category (17/29), compared to *obvious* (7/29) and *cautious* hazards (5/29).

A thematic analysis of hazards missed by both human annotators and the VLM revealed modality-specific blind spots. Across both, missed *obvious* hazards were typically low-salience but visible risks embedded in familiar household elements, such as sharp furniture corners (missed by VLM), exposed plumbing (missed by human), or small choking hazards including paper, carpet strands, or plant material (VLM). Missed *cautious* hazards predominantly involved interaction-dependent risks that required anticipating child behavior, including fall hazards from furniture (missed by both), unsecured or tipping objects (both), breakable glass and mirrors (VLM), and access to unsafe areas such as stairs, doors, or adjacent rooms (both). Finally, missed *latent* hazards reflected risks inferred from environmental context rather than direct visibility, such as hidden electrical outlets and cords (both), cleaning products or alcohol stored out of sight (human), hot liquids in cups (VLM), blind or curtain cords (human), and drowning risks associated with toilets (VLM).

5 Discussion & Future Work

This work examined how VLMs identify child safety hazards in domestic environments and how these interpretations align with human expert judgments. Across all images, we observed substantial overall agreement between human experts and the VLM, suggesting that VLMs can reliably identify many common and salient child safety hazards. At the same time, a non-trivial portion of hazards were missed by either humans or the model, underscoring the inherent difficulty of anticipating risk in dynamic home environments [6, 17].

Agreement was highest for *obvious* hazards, which are directly visible and widely recognized as dangerous. This finding aligns with prior work in automated hazard detection and safety compliance, which consistently reports strong performance for visually salient, object-centric risks [1, 8]. The relatively high agreement for *cautious* hazards further suggests that VLMs can capture some interaction-dependent risks when sufficient contextual cues are present in the image, consistent with emerging work on contextual risk assessment [11]. However, agreement declined for *latent* hazards, reflecting the well-documented challenge of reasoning

about risks that are not directly visible and must be inferred from environmental context [6].

Our disagreement analysis revealed modality-specific blind spots. Human annotators were more likely to miss hazards that were visually unobtrusive, normalized within familiar household settings, or spatially occluded, a pattern consistent with prior findings on inattentive blindness and normalization of risk in everyday environments [12]. In contrast, the VLM more frequently missed hazards that required multi-step reasoning about child behavior such as fall risks from climbing, tipping furniture, or hazards emerging through movement and play. This finding reinforces the idea that child safety hazard anticipation is not merely a perception problem, but a cognitively demanding task shaped by uncertainty, incomplete information, and evolving environments.

Our findings have implications for the design of AI-assisted child safety tools. Rather than replacing caregiver judgment, VLM-based systems may be most effective as complementary aids that surface overlooked risks and prompt reflective safety checks [10]. Their ability to articulate hazards in natural language and reason across different hazard categories highlights potential for explainable, caregiver-facing safety support. At the same time, our results caution against over-reliance on automated systems for hazards requiring nuanced understanding of child behavior or household routines, emphasizing that caregivers should remain actively involved in reviewing and interpreting AI-generated hazard cautions. Furthermore, deployment in safety-critical contexts also raises ethical considerations that require further study.

This work also contributes a conceptual framing of child safety hazards into *obvious*, *cautious*, and *latent* categories. This taxonomy offers a useful lens for understanding why certain hazards are consistently overlooked and may inform future annotation schemes, evaluation benchmarks, and interface design.

This study has limitations that motivate several directions for future work. Our findings are exploratory, based on a small curated sample and a single VLM, and may be influenced by the structured prompt design, and therefore should not be interpreted as evidence of generalizable model performance. Future studies should examine larger and more diverse home environments, incorporate multiple VLMs and prompt variations, and evaluate dynamic scenarios to better capture variability in household layouts, routines, and risk profiles. Longitudinal deployments with caregivers could further clarify how AI-generated hazard interpretations influence awareness, trust, and real-world safety practices over time. In addition, future work may explore fine-tuning VLMs using human annotations and refining prompt design to reduce disagreement in *latent* hazard cases, e.g., through richer contextual prompts, domain-specific training data, or iterative human-in-the-loop refinement. Finally,

future deployments could extend beyond mobile use cases to integrate with home cameras, e.g., baby monitors, and mixed reality interfaces.

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